Interactive Web Authoring

Learning Outcome: To be able to correctly use JavaScript arrays and conditional statements.

You are going to investigate the usage of JavaScript arrays and conditional statements to further improve your understanding of the core concepts of JavaScript.

Create a simple webpage and complete the following tasks:

1. Without testing the below code attempt to correctly estimate the output from the final line.

```
let fruits = ["Apples", "Pear", "Orange"];

// push a new value into the "copy"
let shoppingCart = fruits;
shoppingCart.push("Banana");

// what's in fruits?
alert( fruits.length ); // ?
```

2. Create an array and carry out the following tasks, print the array each time:

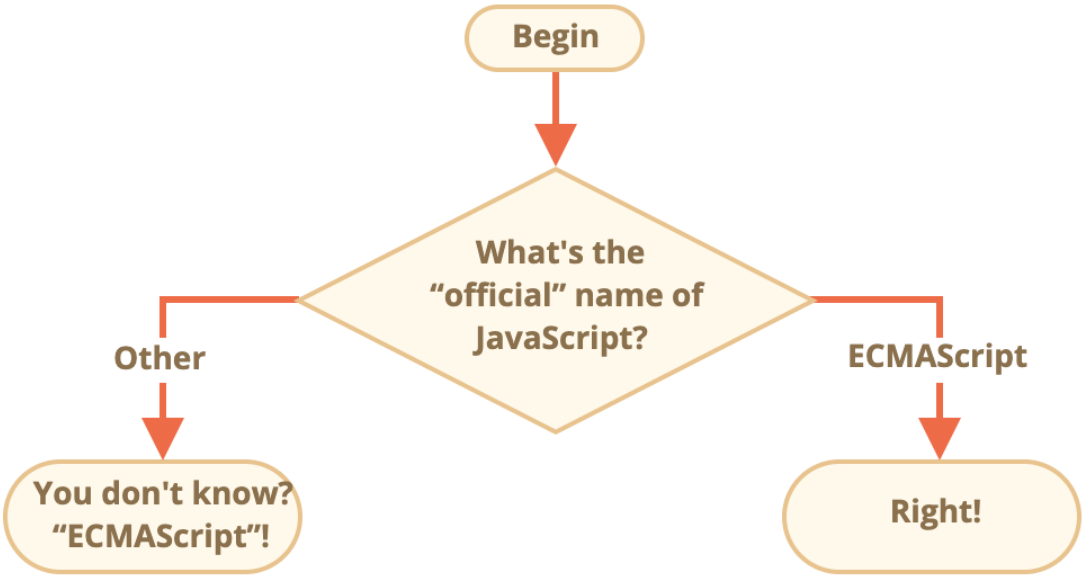
   a. Create an array styles with items “Jazz” and “Blues”.
   b. Append “Rock-n-Roll” to the end.
   c. Replace the value in the middle by “Classics”. Your code for finding the middle value should work for any arrays with odd length.
   d. Strip off the first value of the array and show it.
   e. Prepend Rap and Reggae to the array.

   If this has been completed correctly your results should look like the following:

```
Jazz, Blues
Jazz, Blues, Rock-n-Roll
Jazz, Classics, Rock-n-Roll
Classics, Rock-n-Roll
Rap, Reggae, Classics, Rock-n-Roll
```

3. Using the if..else construct, write the code which asks: ‘What is the “official” name of JavaScript?’

   If the visitor enters “ECMAScript”, then output “Right!”, otherwise – output: “Didn’t know? ECMAScript!” Below is a diagram to aid you in creating the if statement.



4. Using if..else, write the code which gets a number via prompt and then shows in alert:

   - 1, if the value is greater than zero,
   - -1, if less than zero,
   - 0, if equals zero.

In this task we assume that the input is always a number.

Advanced Tasks:

1. Rewrite this if using the ternary operator:

```
let result;

if (a + b < 4) {
  result = 'Below';
} else {
  result = 'Over';
}
```

2. Rewrite if..else using multiple ternary operators. For readability, it's recommended to split the code into multiple lines.

```
let message;

if (login == 'Employee') {
  message = 'Hello';
} else if (login == 'Director') {
  message = 'Greetings';
} else if (login == '') {
  message = 'No login';
} else {
  message = '';
}
```